

Murisphere Workflow Tutorial

Technician, Facility Manager, and Admin Workflow Guide

Murisphere Operations

2026-03-19

Contents

Purpose	2
Roles	2
Training Environment Used In This Guide	2
1. Quick Start and Required Setup	2
1.1 First login and landing dashboard	2
1.2 Card printing preflight	3
1.3 Phone scan verification	3
1.4 Optional desktop companion usage	3
2. Mouse Technician Daily Workflows	3
2.1 Start-of-shift dashboard triage	3
2.2 Print or reprint cage cards	3
2.3 Scan a printed cage card and quick-edit in room	3
2.4 Census and reconciliation	4
2.5 Breeding operations	4
2.6 Health, welfare, mortality, and euthanasia	4
2.7 Tagging, samples, and genotyping logistics	4
2.8 End-of-shift sync discipline	4
3. Facility Manager Daily Workflows	4
3.1 Morning operations dashboard review	4
3.2 Capacity, quota, and chargeback review	4
3.3 Alert ownership and notification dispatch	5
3.4 Recommendation and consolidation workflows	5
3.5 Planner scenarios and supply risk	5
3.6 Desktop companion and office workflows	5
4. Regulatory and Compliance Workflows	5
5. Research Support Workflows	5
6. Visual Walkthrough	6
6.1 Login Screen	6
6.2 Landing Dashboard, Cage Alerts, and Density View	6

6.3 Cage Card Center and Batch Printing	7
6.4 Complete Cage Card	7
6.5 Scan Base URL Setup	8
6.6 Phone Camera Scan of Printed QR	8
6.7 Scan-to-Edit Cage Workflow	9
6.8 Breeding Pair Productivity	9
6.9 Samples and Genotyping Workflow	10
6.10 Recommendations and Outcomes	10
6.11 Planner Scenario Evaluation	11
6.12 Compliance Dashboard	11
6.13 Pedigree Explorer	12
6.14 Scan Troubleshooting Reference	12
7. End-of-Shift and End-of-Day Checklists	13
Technician	13
Facility Manager / Admin	13
8. Troubleshooting and Escalation	13

Purpose

This tutorial is organized by **real vivarium workflows** so technicians, PIs, and facility managers can move through daily work quickly, accurately, and consistently.

Murisphere is built around one operating rule:

Print the cage card -> scan the QR with a phone -> open the cage in the browser -> complete the task immediately.

Roles

- Technician
- PI / Facility Manager
- Admin

Training Environment Used In This Guide

The sample screenshots and examples in this tutorial assume a realistic demo environment:

- 20 labs
- 3,000 cages
- multiple projects per lab
- variable lab sizes (small, medium, large)
- active alerts, planner scenarios, breeding pairs, samples, and compliance records

1. Quick Start and Required Setup

1.1 First login and landing dashboard

1. Log in from phone, tablet, laptop browser, or the optional desktop companion.
2. After login, review the landing dashboard first.

3. Check alert severity, overdue work, room pressure, and planner risk before starting rounds.
4. Use clickable cage, project, and lab names to open the detailed view directly.

1.2 Card printing preflight

1. Open the cage card center.
2. Confirm the **Scan Base** URL points to a phone-reachable host or domain.
3. Print cards at **100% scale**.
4. Use the QR square for phone scanning.
5. Treat the 1D barcode strip as optional hardware-scanner support only.

1.3 Phone scan verification

1. Use the phone camera, not a separate mobile app.
2. Scan the printed QR code on a real card.
3. Tap the browser prompt.
4. Confirm cage ID and location before editing.
5. Save one test update and verify it appears immediately in cage history.

1.4 Optional desktop companion usage

Use the Tauri desktop companion for: - batch card printing - reporting and exports - office workstations - managed desktop environments

Use the web workflow for: - phone scanning in rooms - quick cage edits - tablet rounds - anywhere zero-install access matters

2. Mouse Technician Daily Workflows

2.1 Start-of-shift dashboard triage

1. Open the dashboard.
2. Review high-severity alerts first.
3. Review overdue tasks and protocol blocks.
4. Prioritize rooms with dense alert clusters or urgent welfare events.

2.2 Print or reprint cage cards

1. Search for the target cages.
2. Open the card center.
3. Batch-print selected cards.
4. Inspect print quality before placing cards in service.
5. Verify the QR is sharp, square, and fully visible.

2.3 Scan a printed cage card and quick-edit in room

1. Scan the printed QR with the phone camera.
2. Open the cage record in the browser.
3. Verify cage code, room, rack, and breeding status.
4. Update counts, notes, litter events, or status.
5. Save and confirm the audit trail entry appears.

2.4 Census and reconciliation

1. Start a census session for the room.
2. Scan each cage in sequence.
3. Reconcile observed M/F counts with the system totals.
4. Record mismatch notes before leaving the room.
5. Complete the session and verify all scans are synced.

2.5 Breeding operations

1. Create or review breeding pairs.
2. Record timed mating, plug check, litter, weaning, and transfer events.
3. Review pair productivity before keeping older breeders active.
4. Retire non-productive pairs when criteria are met.

2.6 Health, welfare, mortality, and euthanasia

1. Record welfare concerns during rounds.
2. Open vet cases and treatment schedules when needed.
3. Record mortality with necropsy requirement if appropriate.
4. Record euthanasia method, reason, and disposition.
5. Confirm critical welfare events raise alerts.

2.7 Tagging, samples, and genotyping logistics

1. Add ear tag, chip, tube, or well identifiers to animals.
2. Create sample records with unique sample codes.
3. Advance sample state through collection, shipment, receipt, and result.
4. Confirm chain-of-custody events are complete before handoff.

2.8 End-of-shift sync discipline

1. Confirm offline or queued writes are fully synced.
2. Close open census sessions and unfinished round notes.
3. Escalate unresolved high-severity findings before leaving.

3. Facility Manager Daily Workflows

3.1 Morning operations dashboard review

1. Open the landing dashboard.
2. Review alert counts by severity and category.
3. Review room density, cage pressure, and open task volume.
4. Open high-risk cages directly from dashboard links.

3.2 Capacity, quota, and chargeback review

1. Review facility capacity and room/rack utilization.
2. Review lab quotas and projected cage load.
3. Review chargeback and per-diem outputs for the current billing window.
4. Investigate overloaded rooms or underutilized allocations.

3.3 Alert ownership and notification dispatch

1. Review active alerts.
2. Acknowledge items that are known and assigned.
3. Dispatch notifications for unresolved urgent items.
4. Confirm escalation channels are configured correctly.

3.4 Recommendation and consolidation workflows

1. Generate recommendations.
2. Review each item with cage, lab, and risk context.
3. Decide: accept, adjust, ignore, or complete.
4. Track outcomes so the system can show operational value over time.

3.5 Planner scenarios and supply risk

1. Create planner scenarios with demand targets and needed-by dates.
2. Attach project-level demand where required.
3. Evaluate projected animal deficits, cage demand, and risk level.
4. Assign action owners for high-risk scenarios.

3.6 Desktop companion and office workflows

Use the optional desktop companion for: - high-volume cage card printing - long report review sessions - export administration - office or front-desk workstation use

4. Regulatory and Compliance Workflows

1. Monitor protocol expiration and hard-stop blocked edits.
2. Record deviations, CAPA items, and closure state.
3. Track quarantine intake, hold, release, and overdue cases.
4. Track qualification requirements for assigned procedures.
5. Capture attachments and e-signature evidence where required.
6. Use audit history to answer who changed what and when.

5. Research Support Workflows

1. Explore pedigree trees for breeding decisions.
2. Submit genotyping orders and receive provider callbacks.
3. Review Mendelian ratio and genotype deviation alerts.
4. Review planner scenarios tied to project demand.
5. Export CSV, PDF, and workflow job outputs for downstream systems.

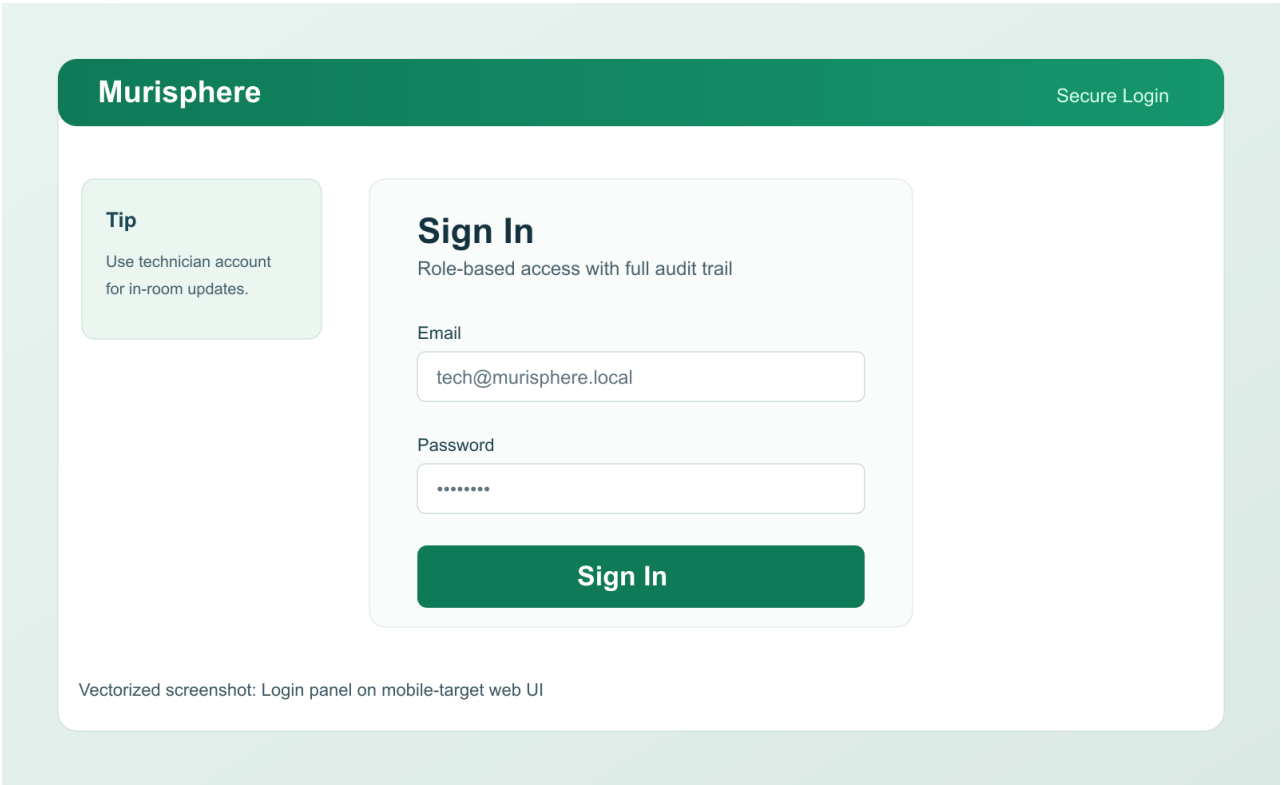
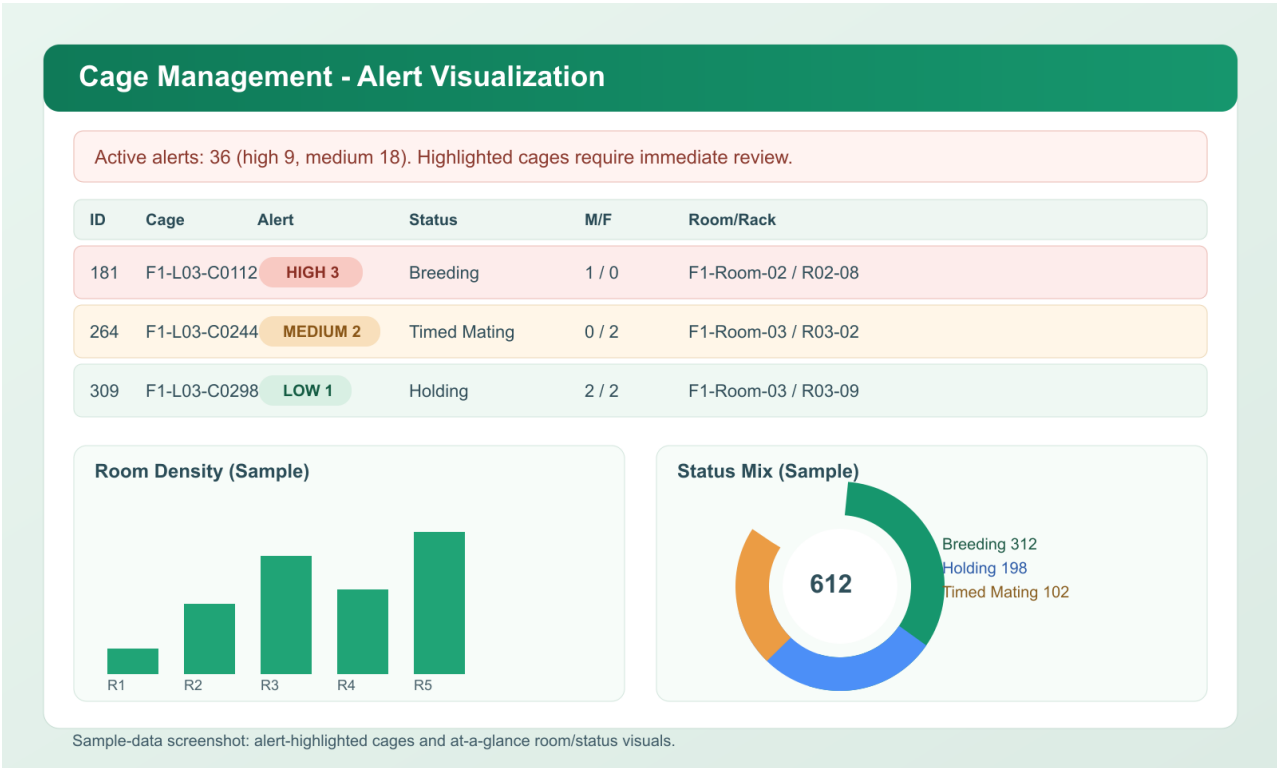


Figure 1: Login screen with role-aware access controls

6. Visual Walkthrough

6.1 Login Screen

6.2 Landing Dashboard, Cage Alerts, and Density View



Use

this screen immediately after login to prioritize work.

6.3 Cage Card Center and Batch Printing

Cages > Generate + Print

Generate Cards

Quick Print

Card Preview (C-A1-0243)

QR code (scan with phone)

Barcode (optional)

Scan URL: https://vivarium.example.org/scan/tok_a1_0243

Printing Checklist

1. Confirm Scan Base URL is phone-reachable

2. Confirm each card shows visible QR square

3. Print at 100% scale

4. Test one printed card with phone camera

Vectorized screenshot: Cage card generation and print readiness

Use

this workflow to generate or reprint cards before rounds.

6.4 Complete Cage Card

Murisphere Cage Card (Rendered Preview)

Designed for fast room-floor scanning, complete husbandry context, and compliance readiness

Cage: F1-L03-C0215

Group Owner: Dr. Meosha Hudson

Group Name: Hudson Lab

Projects: L03-PRJ-01, L03-PRJ-05

Location: Room 1, Breeding Rack 1, Slot 6A

Protocol: IACUC-2026-0312

Description: PdxCre longitudinal cohort

Breeding: Pairing 35 (Timed Mating)

Protocol Expires: 12/31/2026

Population: M1 / F2 / T3

Tracked IDs shown: 3 of 3

Scan with phone camera

Barcode (optional)

ID	Sex	DOB	Genotype	Status
258	M	10/10/2024	PdxCre Tg/Azelia	Breeder
1L	F	10/28/2024	PdxCre +/-tg	Breeder
1L1R	F	10/28/2024	PdxCre +/-	Breeder

Litters

#	DOB	Born	Survived	M/F	DoW
1	11/12/2024	3	3	2/1	01/15/2025

#	DOB	Born	Survived	M/F	DoW
2	01/08/2025	5	5	1/4	02/05/2025

Pop-

ulation is the full cage total (M/F/T). Tracked IDs Listed shows how many individual records are

printed. Litters include DoW (date of weaning).

6.5 Scan Base URL Setup

Set Scan Base URL Correctly

Scan Base URL

http://10.0.0.24:8000

Use This

LAN IP or production domain
Reachable by phone browser
Same network as room devices

Avoid This

http://localhost:8000
http://127.0.0.1:8000
Phone cannot reach your laptop loopback

Never print cards with localhost or another non-phone-reachable address.

6.6 Phone Camera Scan of Printed QR

Phone Scan Workflow

Printed Cage Card

Cage ID: C-A1-0243
Strain: C57BL/6J
M/F: 2 / 3

Barcode (optional)

QR (scan with phone camera)

Open Link

Cage C-A1-0243
Status: Holding
M/F: 2 / 3

Open Murisphere

Key Action

1. Camera detects QR
2. Tap link banner
3. Verify cage ID
4. Open for edits

phone camera should open the browser directly from the printed QR.

The

6.7 Scan-to-Edit Cage Workflow

Scan/Edit Workflow

1. Enter QR token or scan link

Open Cage

C-A1-0243 (lookup 132ms)

C57BL/6J | WT/WT | Breeding

Room A1 / Rack R3

Male Count Female Count

Save Changes

2. Phone Camera Scan

Open in Browser

Tap banner to open /scan/tok_a1_0243...

Vectorized screenshot: Scan token lookup and edit confirmation

Use

this flow for in-room edits with minimal taps.

6.8 Breeding Pair Productivity

Breeding Pair Workflow

Create Pair (Sample Data)

Cage: F1-L03-C0244

Sire: PAIR-SIRE-001 (M, fl/+)

Dam: PAIR-DAM-001 (F, +/-)

Started: 2026-03-05

ACTIVE PAUSE RETIRE

Pair Productivity (Sample)

Pair-12

Pair-19

Pair-24

Pair-28

Litters: 4 | Avg Survived: 5.2 | Total Survived: 21

Breeding Events Timeline (Sample)

03/05 Setup

03/08 Plug Check

03/28 Weaning

04/01 Retire

04/03 New Pair

Sample-data screenshot: pair creation, status transitions, and productivity trend.

Review productivity before continuing or retiring a pair.

6.9 Samples and Genotyping Workflow

Sample Tracking + Genotyping Orders

Sample Lifecycle (Animal PAIR-SIRE-001)

COLLECTED

2026-03-05 09:12

SHIPPED

Tracking: TNX-334512

RECEIVED

Provider: Transnetyx

RESULTED

Result: fl/+ (Cre Panel)

Genotyping Order (Sample)

OrderRef: GENO-20260305-7fa2b1
Provider: Transnetyx | Status: submitted

Item	SampleCode	Animal	Result
1	SMP-001	PAIR-SIRE-001	fl/+
2	SMP-002	PAIR-DAM-001	+/+

Callback Ingestion Result (Sample)

Provider callback authenticated via X-Provider-Token.
2 animals updated and genotype history appended.

updatedAnimals: 2

status: received

Sample-data screenshot: chain-of-custody and provider callback result flow.

Sample and callback history should be complete and traceable.

6.10 Recommendations and Outcomes

Recommendation Lifecycle

Generated Recommendations (Sample)

Open: 14 | Accepted: 9 | Adjusted: 3 | Ignored: 2 | Completed: 7

Rec #381 - consolidate_cage

Cage F1-L03-C0112 has low density (1 animal) and compatible strain/genotype group nearby.
Suggested target: F1-L03-C0120 | Estimated cage reduction: 1

ACCEPT

ADJUST

IGNORE

COMPLETE

Decision Outcome Trend (Sample)

Day	Outcome
Mon	Low
Tue	Medium
Wed	Medium
Thu	Low
Fri	High

Action Notes (Sample)

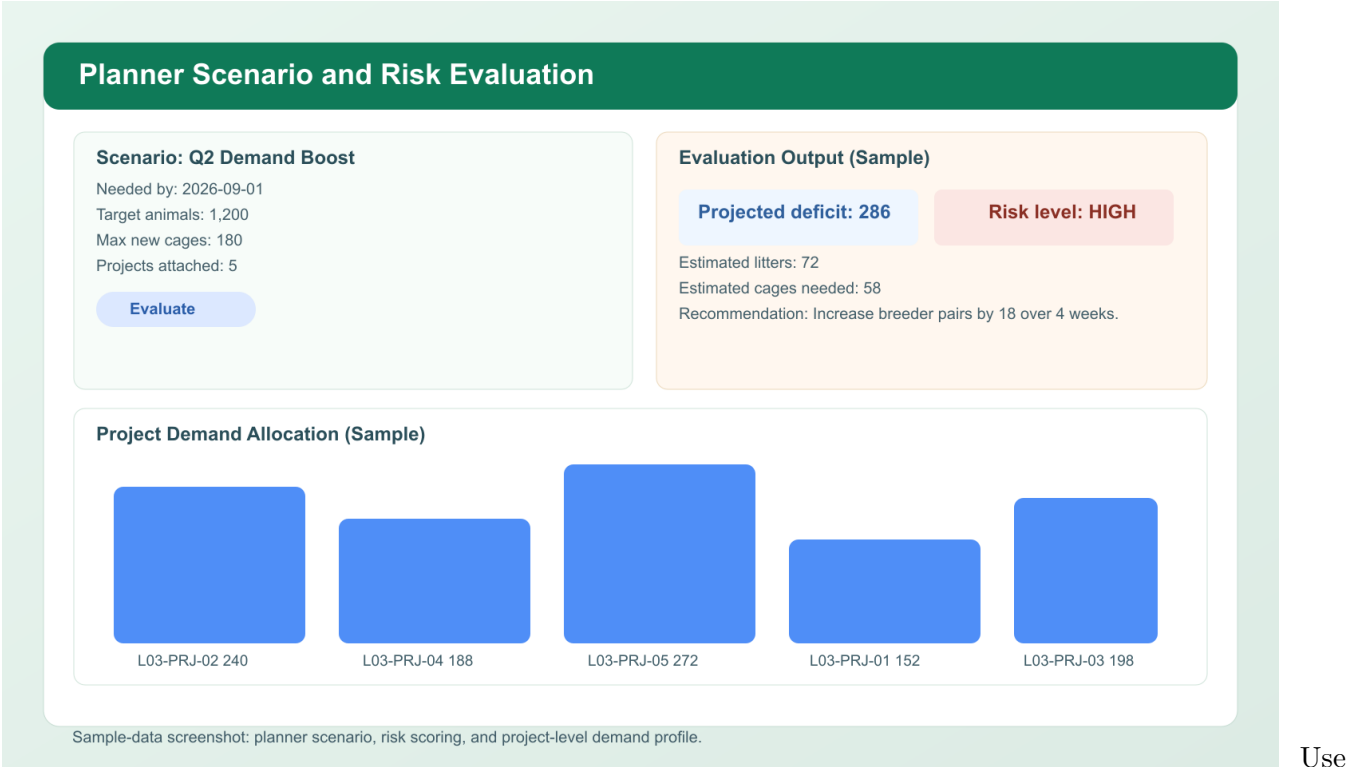
Adjusted: "Delay consolidation until post-wean on 2026-03-28"
Accepted by: Admin User at 2026-03-05 14:09
Expected outcome: reduce 12 cages this month.

Sample-data screenshot: decisionable recommendations with measurable outcomes.

Recommendations should become assigned actions, not passive dashboards.

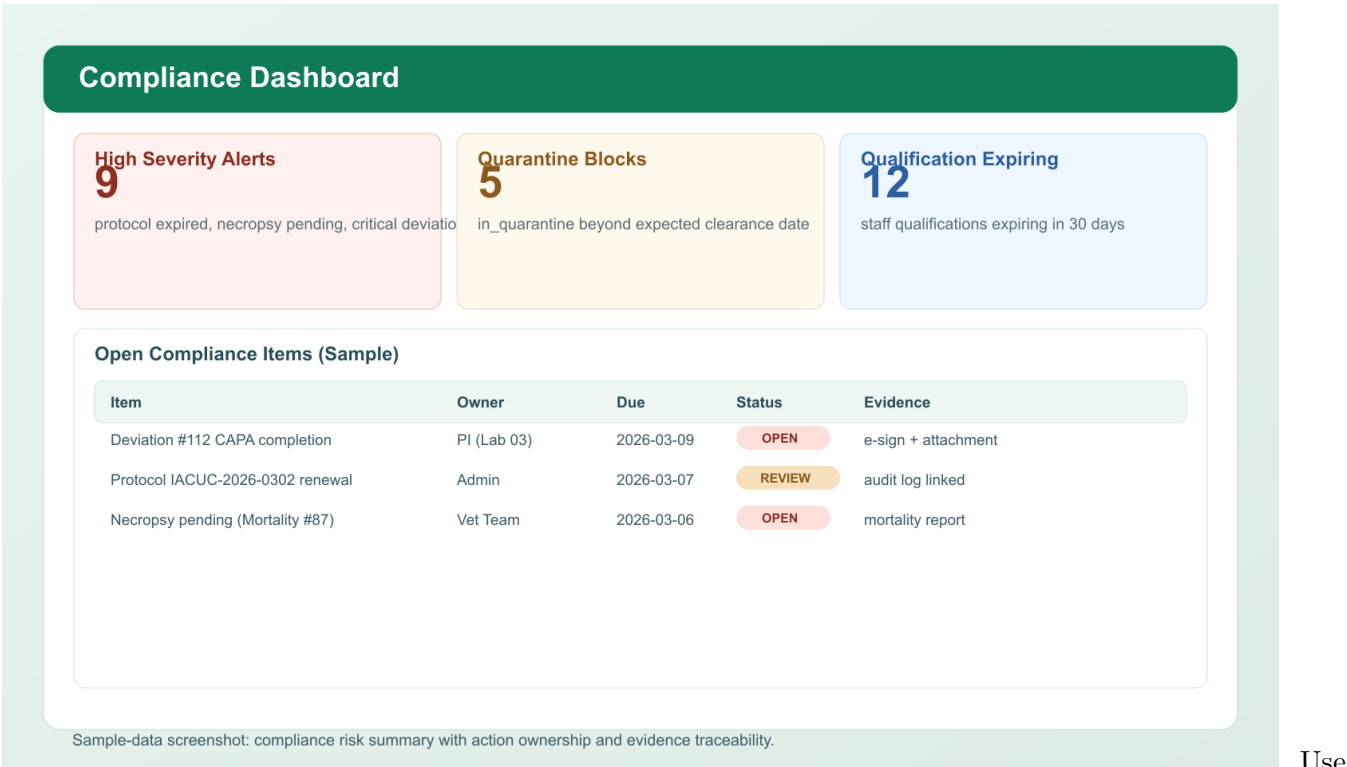
10

6.11 Planner Scenario Evaluation



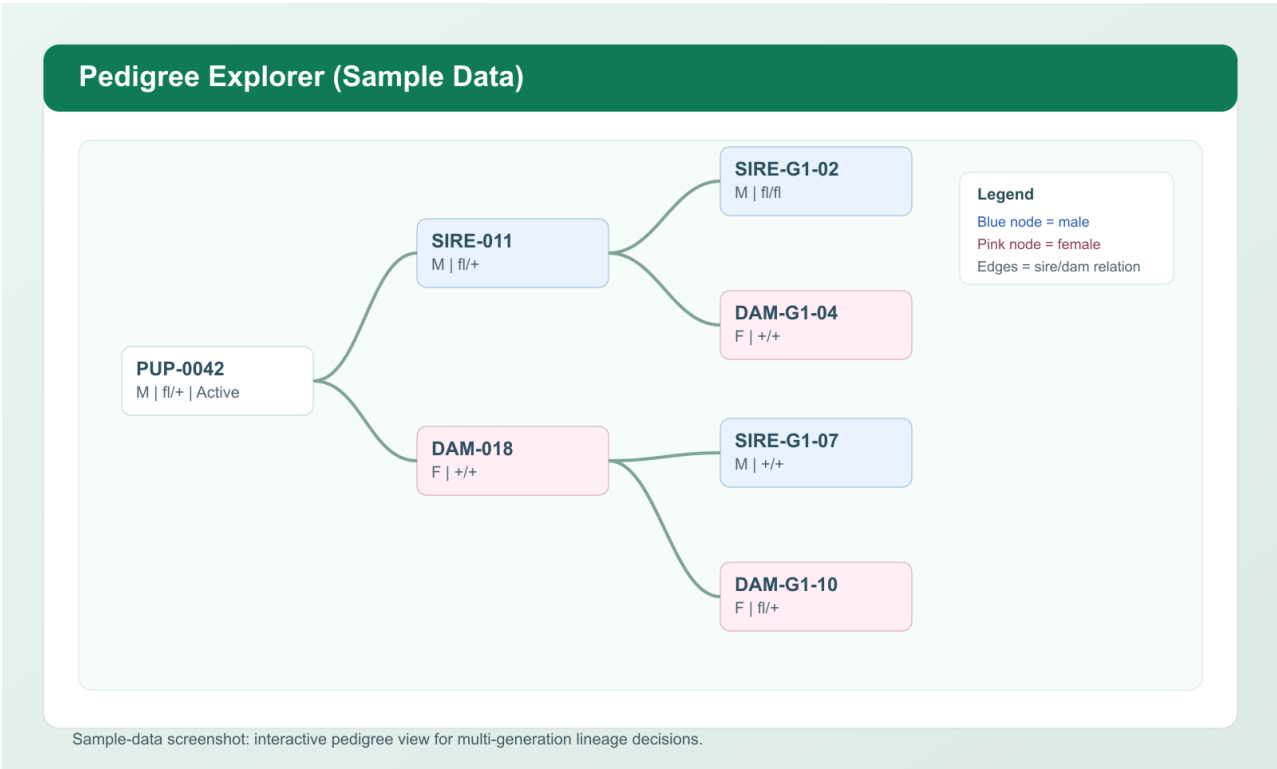
Use the planner to understand supply gaps before they become operational failures.

6.12 Compliance Dashboard



Use this view for protocol, deviation, and evidence review.

6.13 Pedigree Explorer



Pedigree visualization is useful for breeding strategy and genotype interpretation.

6.14 Scan Troubleshooting Reference

Scan Troubleshooting

Problem

Blank / broken QR
No scan prompt
Wrong URL opens
Link opens slowly

Cause

Cached JS / blocked CDN
Low contrast / glare
Scan Base URL mis-set
Weak room Wi-Fi/LAN

Action

Hard refresh + regenerate cards
Improve lighting and distance
Set phone-reachable base URL
Retest on same network

If unresolved, capture browser console errors and include the cage card ID and timestamp.

Keep this guidance available anywhere cards are printed.

7. End-of-Shift and End-of-Day Checklists

Technician

1. Confirm all queued writes are synced.
2. Confirm mortality, necropsy, euthanasia, and vet notes are complete.
3. Confirm all required room sessions are closed.
4. Hand off unresolved urgent issues before leaving.

Facility Manager / Admin

1. Clear or owner-assign all high-severity alerts.
2. Review unresolved deviations and quarantine holds.
3. Review planner and recommendation items that remain high risk.
4. Export required summaries for audit, PI review, or operational handoff.

8. Troubleshooting and Escalation

- **No scan prompt appears:** improve lighting, flatten the card, and try the QR square again.
- **The phone reads only the barcode strip:** scan the QR square, not the 1D barcode.
- **Wrong page opens:** correct the Scan Base URL and reprint the card.
- **QR appears blank or broken:** verify the QR asset endpoint and regenerate the card.
- **Save appears queued:** reconnect and confirm sync before leaving the room.
- **Protocol hard-stop prevents edit:** assign a valid active protocol first.
- **Callback or sample mismatch occurs:** verify callback token, sample code, and animal mapping.
- **Planner risk remains high:** assign an owner and create a corrective action immediately.